

INTERMEDIATE MACROECONOMICS
SOLOWIAN MODEL OF GROWTH
29. BALANCED GROWTH

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LAW OF MOTION OF CAPITAL PER EFFECTIVE WORKER

- using the definition of growth rate g_k :
 - $k(t+1) - k(t) = g_k \times k(t)$
- first step: compute g_k
 - since $k = K / AN$
 - then $g_k = g_K - (g_A + g_N)$
- hence $k(t+1) - k(t) = [g_K - (g_A + g_N)] \times k(t)$
- which implies: $k(t+1) - k(t) = g_K \times k(t) - [g_A + g_N] \times k(t)$

GROWTH RATE OF CAPITAL

- evolution of the capital stock is driven by investment and depreciation: capital tomorrow = capital today + investment today – depreciation today
 - $K(t+1) - K(t) = I(t) - \delta \times K(t)$
- growth rate of capital:
 - $g_K = [K(t+1) - K(t)] / K(t) = [I(t) / K(t)] - \delta$
- since $K(t) = k(t) \times A(t)N(t)$, we conclude that
 - $g_K \times k(t) = I(t) / [A(t)N(t)] - \delta \times k(t) = i(t) - \delta \times k(t)$

BACK TO LAW OF MOTION OF CAPITAL PER EFFECTIVE WORKER

- we have:

$$1. \quad k(t+1) - k(t) = g_K \times k(t) - [g_A + g_N] \times k(t)$$

$$2. \quad g_K \times k(t) = i(t) - \delta \times k(t)$$

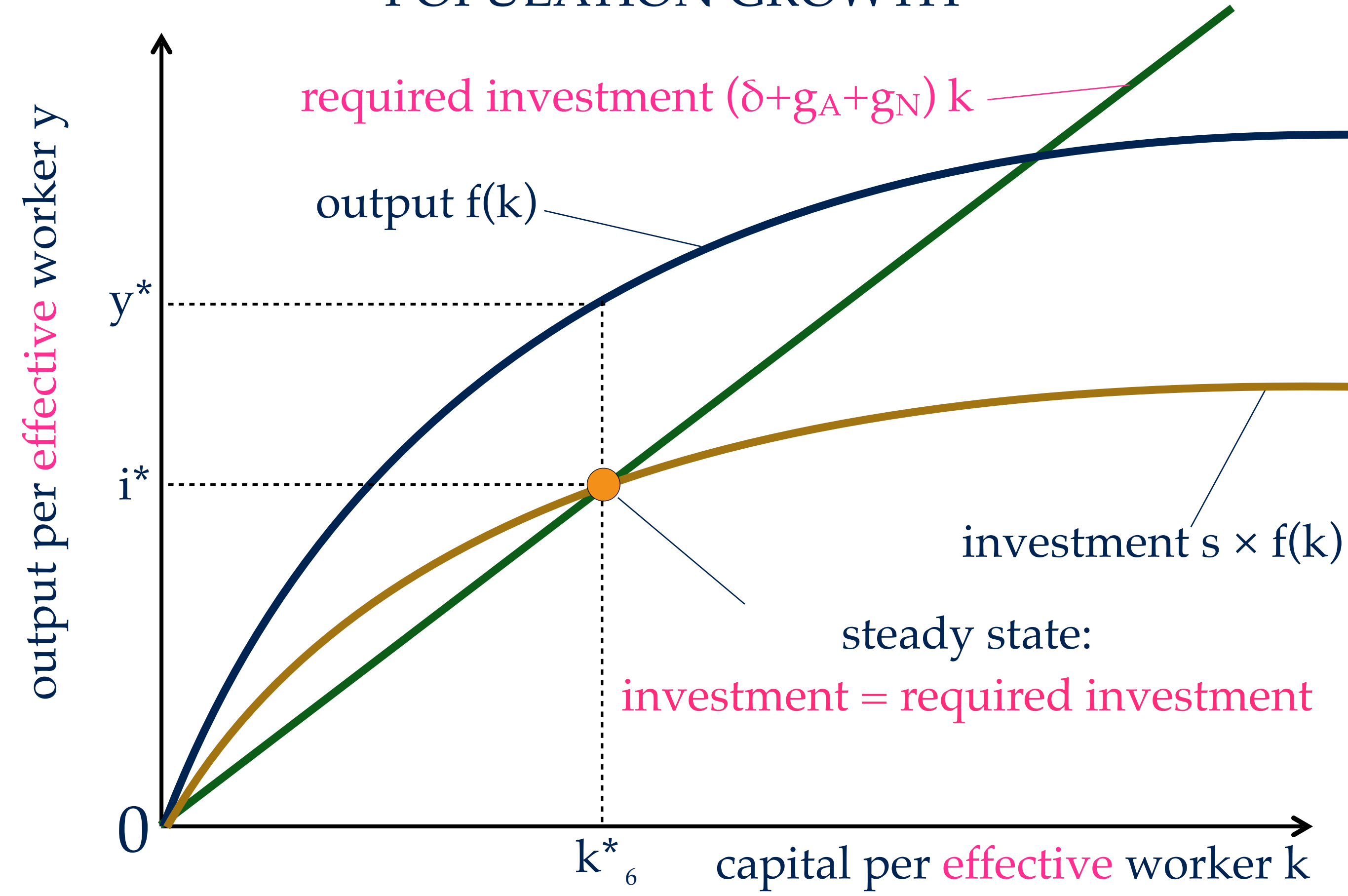
$$3. \quad i(t) = s \times f(k(t))$$

- hence: $k(t+1) - k(t) = s \times f(k(t)) - [\delta + g_A + g_N] \times k(t)$
- same law of motion as in basic Solow model
- but δ is replaced by $\delta + g_A + g_N$

THE STEADY STATE

- capital per effective worker is constant
- output per effective worker is constant
- using the law of motion of capital per effective worker, we find that steady-state capital per effective worker k^* satisfies
 - $s \times f(k^*) = [\delta + g_A + g_N] \times k^*$
- to maintain $k = K / AN$ constant, there must be enough investment
 - to cover depreciation of K (δ)
 - to cover growth of A (g_A) and growth of N (g_N)

EQUILIBRIUM WITH TECHNOLOGICAL & POPULATION GROWTH



BALANCED GROWTH IN STEADY STATE

definition of steady state

k^*

y^*

	Growth Rate:
Capital per effective worker	0
Output per effective worker	0
Capital per worker	g_A
Output per worker	g_A
Labor	g_N
Capital	$g_A + g_N$
Output	$g_A + g_N$

BALANCED GROWTH IN STEADY STATE

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growth rate of population is given

BALANCED GROWTH IN STEADY STATE

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$$(K/N)^* = k^* \times A$$

$$(Y/N)^* = y^* \times A$$

there is **balanced growth** because several variables grow at the same rate

BALANCED GROWTH IN STEADY STATE

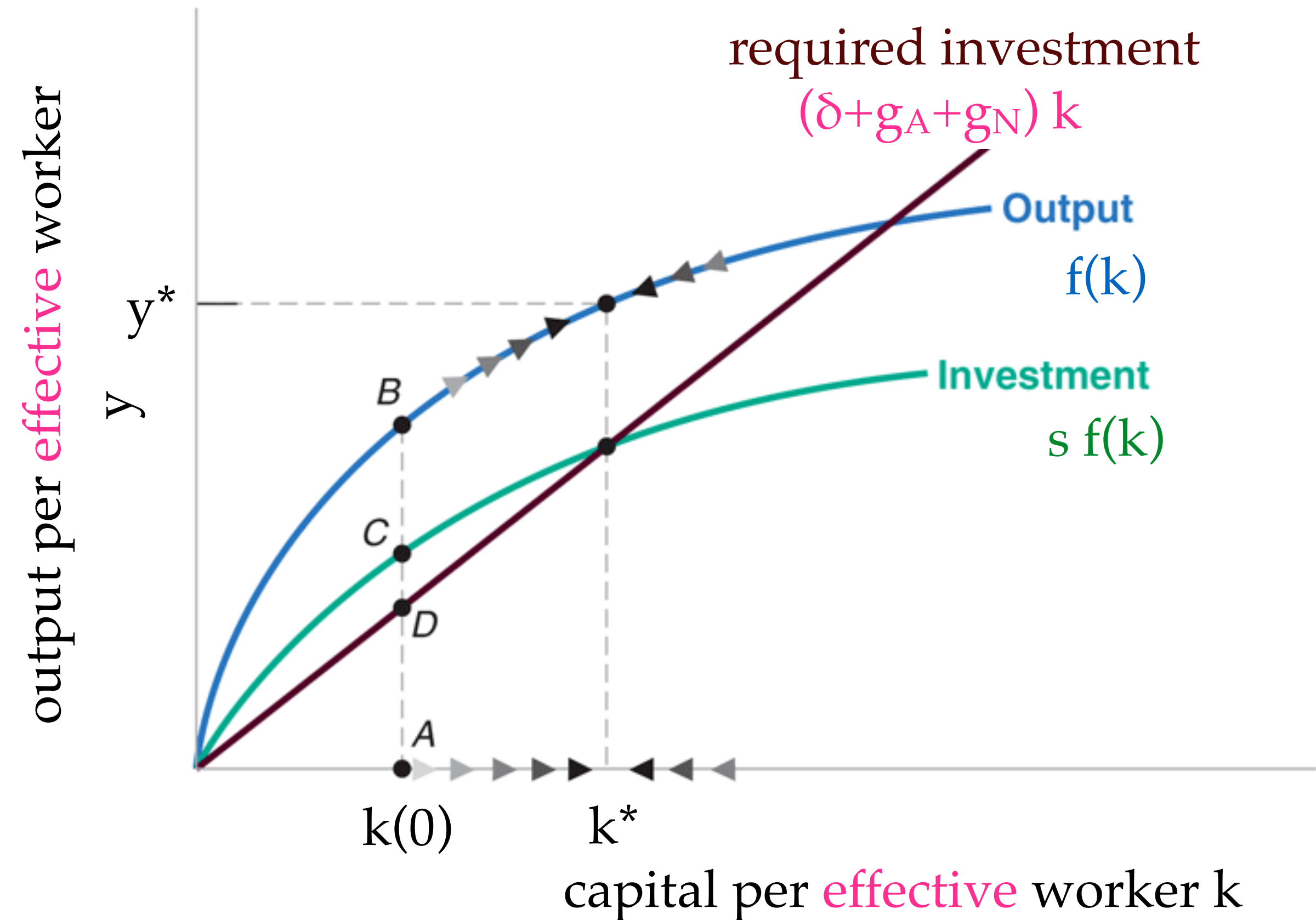
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$$K^* = k^* \times A \times N$$

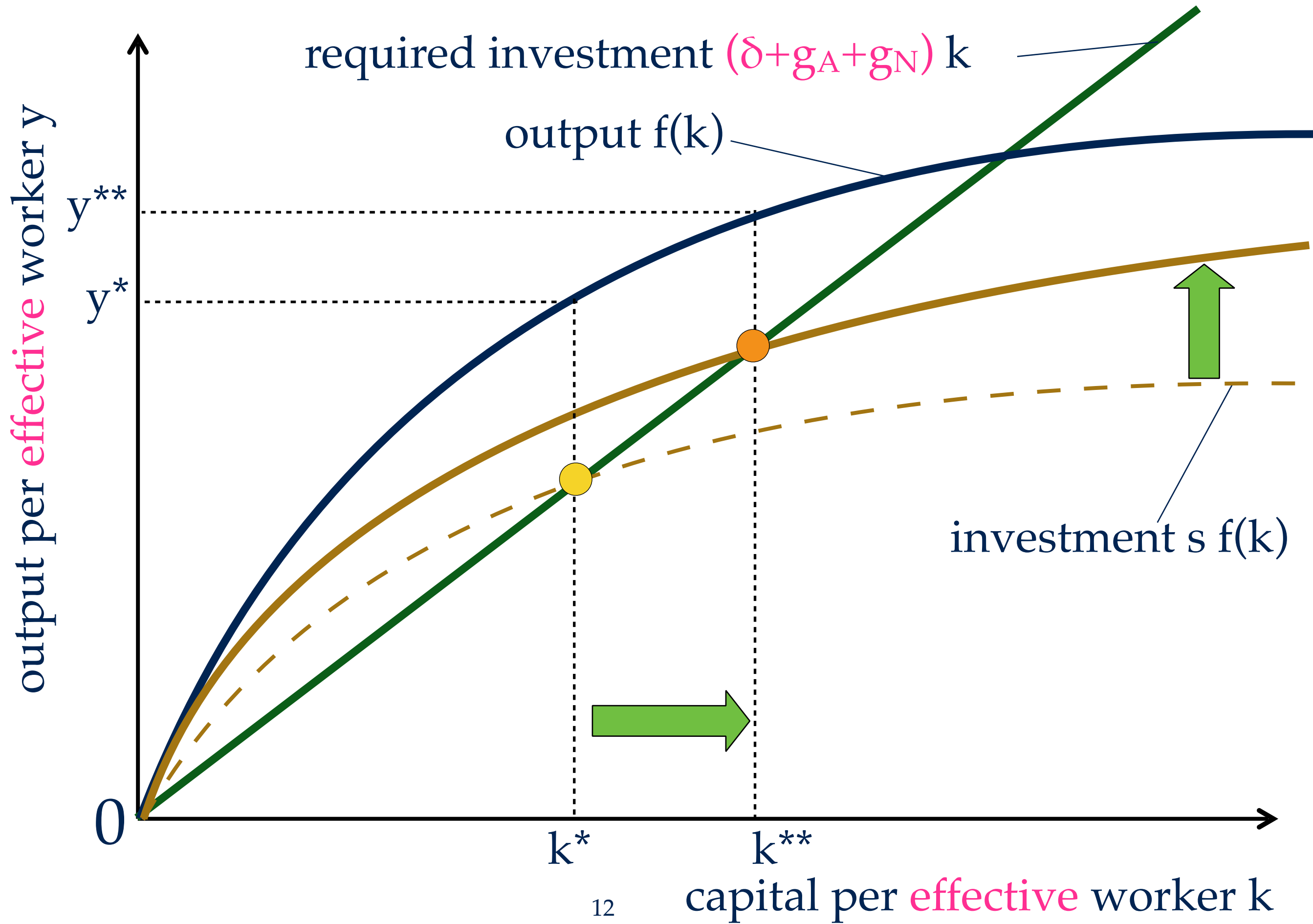
$$Y^* = y^* \times A \times N$$

there is **balanced growth** because
several variables grow at the same rate

EQUILIBRIUM DIAGRAM



INCREASE IN SAVING RATE: STEADY STATE



INCREASE IN THE SAVING RATE: DYNAMICS

